Germany's Redispatch Is a Network Problem, Not a Carbon Saving

Reading Germany's 2025 Congestion-Management Data Through a Carbon Lens

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1. From the Post to the Data

A few weeks ago I posted about Germany's 2024 grid-congestion bill: €2.9 billion, and 9.4 TWh of renewable generation curtailed after it had been produced. The post included a line from one of my MSc students that stuck with me: "The €2.9 billion is not the cost of building the wrong grid. It's the annual cost of not building the right one." In my first write-up from that digging I showed that a national supply-and-demand indicator misses about 17 times the real redispatch volume. This note asks the two questions that came next. Why does the national view miss it so badly? And when renewable output is curtailed for network reasons, does the country actually save any carbon?

2. Data and Method

I combined 8,760 hourly observations for 2025 from SMARD (grid load, residual load, generation by technology) with the monthly redispatch series published through SMARD and the Bundesnetzagentur: reductions by technology, reductions by plant type, compensation by recipient and congestion-management cost by instrument. Hourly carbon intensity is estimated from generation by fuel and the emission factors lignite 1,020, hard coal 820, other conventional 700, fossil gas 490 and biomass 200 gCO_2/kWh, which gives a 2025 system total of 138.5 Mt CO_2.

Negative residual load, meaning wind plus solar output above grid load, is used as a national potential-surplus indicator. It is not a measure of curtailment, and the point of this note is partly that the two are different things.

3. Redispatch Does Not Follow the National Balance

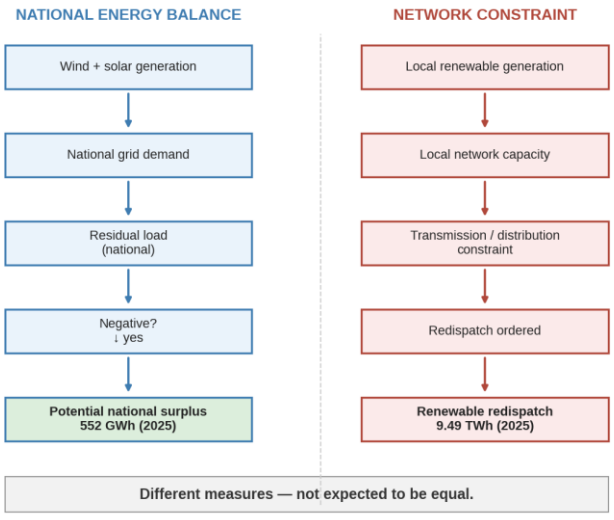


Figure 1. Two different measures: the national energy balance and the local network constraint.

The national indicator asks whether wind and solar exceeded total demand in an hour. Redispatch is ordered because a particular line or substation is overloaded, which depends on where generation and load sit and not on the national total. Those are different questions and there is no reason for the answers to match. They do not. The indicator shows 202 surplus hours in 2025, worth 552 GWh. Renewable redispatch reductions were 9,488 GWh, about 17 times larger. In October the gap was close to 29-fold (44 GWh of potential surplus against 1,252 GWh of redispatch); in June, the month with the most surplus hours, it narrowed to about 4-fold. Monthly correlation between the two series is only r = 0.42.

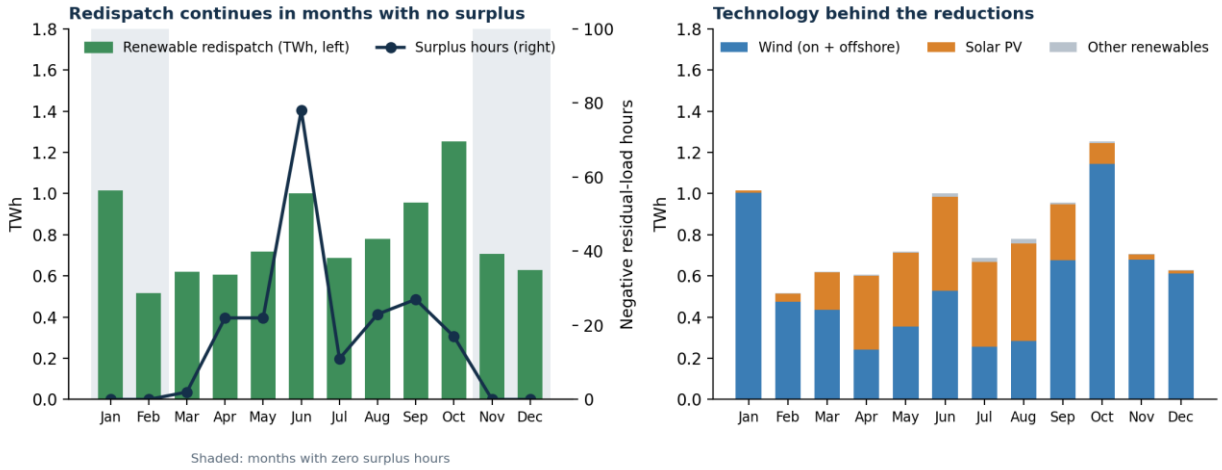


Figure 2. Monthly renewable redispatch against surplus hours (left) and technology behind the renewable reductions (right), 2025.

The clearest evidence is in the winter. January, February, November and December have no surplus hours at all, and still carried between 516 and 1,016 GWh of renewable redispatch each month (left panel of Figure 2). A national balance view would call those months trouble-free.

4. Wind Drives Redispatch, Solar Drives the Surplus

The technology split makes the same point from another side. Wind accounts for about 70% of renewable redispatch reductions in 2025 (offshore 35%, onshore 35%) and PV for about 29%. The surplus hours, on the other hand, are solar hours: during them PV averages roughly 36 GW. So the indicator is largely a measure of midday solar, while the redispatch volume is largely a measure of northern wind meeting a stretched transmission corridor.

SMARD's quarterly reporting shows a second split. In Q2 2025, 49% of renewable redispatch was caused by distribution-grid constraints and 51% by transmission, and of the 2.3 TWh of renewable redispatch that quarter, PV accounted for about 72% of the volume caused at distribution level. By Q3 the split had moved to 41% distribution and 59% transmission. In plain terms there are at least two distinct problems under one label: summer solar behind local distribution constraints, and winter wind behind transmission bottlenecks. A single national number cannot represent either.

5. More Conventional Capacity Than Renewables Is Redispatched Down

The public discussion of curtailment is mostly about renewables. The plant-type series shows a different picture. In 2025 conventional plants were reduced by 13.9 TWh against 9.5 TWh for renewable sources, about 1.5 times as much. Conventional reductions exceeded renewable ones in nine of twelve months; the

exceptions are May, August and October. From November to February the ratio is 2 to 3 times (January 2.1, February 3.2, November 2.5, December 2.6).

Table 1. Redispatch volume and compensation by recipient, 2025.

	Reductions (TWh)	Compensation (€ m)
Conventional plants	13.9	930
Renewable sources	9.5	435
Balance-responsible parties	n/a	241
Total congestion-management cost (all instruments)		3,079

Compensation follows the same pattern: conventional plants received more than twice as much as renewable sources (€930 million against €435 million). I have not divided one by the other to get a price per MWh. The reduction series covers downward adjustments only, while compensation is booked by recipient, so the ratio would not be similar.

6. Shifted Carbon, Not Avoided Carbon

The carbon question is whether curtailing renewables for network reasons is offset by what replaces them. SMARD's own quarterly commentary says it is not offset in a helpful direction: the missing renewable energy is made up mainly by gas and hard-coal plants. Taking that at face value, I applied the same emission factors to the backfill. A 51/49 gas and hard-coal mix works out at 651 gCO_2/kWh, and applying it to 9.49 TWh of renewable redispatch adds about **6.2 Mt CO_2**, 4.5% of the 138.5 Mt system total. Using the blended fossil average of 771 gCO_2/kWh instead gives about **7.3 Mt**, or 5.3%.

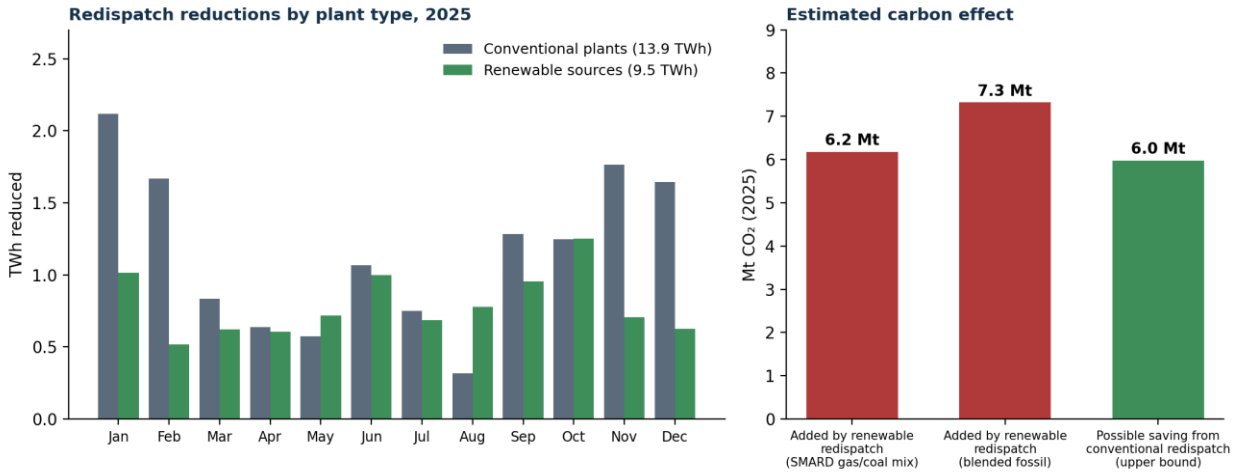


Figure 3. Redispatch reductions by plant type, by month (left), and estimated carbon effect (right), 2025.

The other side of the ledger is the conventional redispatch. If every reduced conventional plant were a coal or lignite unit (about 920 gCO_2/kWh) displaced by gas (490), the 13.9 TWh would save up to 6.0 Mt, 4.3% of the total. That is an upper bound built on an assumption, because no published series tells us which technologies replace which. It is shown as the best case, not as a result.

Put the two together and the best-supported reading is close to a wash: the net national carbon effect of redispatch is close to zero on the evidence available, and if it is a saving at all, it comes from the fuel swaps

on the conventional side, not from curtailing renewables. For renewable curtailment specifically, the evidence points to carbon that is moved to other plants, not carbon that is avoided.

7. What This Means for Carbon Accounting and Decarbonisation Analytics

- **Do not treat curtailment as a national-surplus story.** A supply-demand indicator is a reasonable screen for storage or flexibility needs. As a measure of curtailment it is off by an order of magnitude. Locational data is needed, and the available monthly series cannot supply it.
- **Flexibility has to be placed as well as timed.** A load that soaks up midday solar in the south does nothing for northern wind behind a transmission bottleneck. Siting and grid-connection location matter at least as much as the hour.
- **Match claims to what physically arrived.** Hourly grid carbon intensity built from realised generation already includes the backfill from curtailment. A market-based claim tied to one specific wind farm may not: the matched volume on paper can include output that was curtailed and replaced by fossil plant. That is a boundary question worth checking in 24/7 matching schemes.
- **Read the cost with the carbon.** About €3 billion a year buys a carbon outcome that, on this evidence, is roughly neutral. The case for grid investment rests more strongly on cost and on the 9–10 TWh of clean energy not delivered than on avoided emissions from redispatch itself.

8. Limitations

The redispatch data is monthly and by technology or plant type; there is no hourly, regional or network-element resolution, so nothing here identifies a specific constraint. The backfill mix comes from SMARD's quarterly commentary and I have applied it to the whole year. The renewable-side carbon estimate is therefore an estimate with an evidence base, while the conventional-side saving is an assumed upper bound. Emission factors are representative, not plant-specific. The comparison ignores cross-border flows and any effect on fuel or carbon prices. The year is 2025 only.

9. Conclusion

Germany's redispatch volume is set by where the network is stretched, not by how much wind and solar there is nationally. That is why it continues through winter months with no surplus at all, why wind and not solar dominates it, and why a national indicator misses it by a factor of 17. It is also why curtailing renewables there does not translate into a clean national saving: roughly 6 to 7 Mt of CO_2 is moved onto gas and hard coal, about the same order as the most optimistic saving on the conventional side. The practical lesson for anyone doing carbon or energy analytics is the one from the first paper: check what a headline number is averaging over before drawing a conclusion from it.

Methodology, hourly dataset and Excel workbook available on request. Data: SMARD (Bundesnetzagentur) hourly consumption and generation, 2025; Bundesnetzagentur and SMARD redispatch, compensation and congestion-management series; SMARD quarterly congestion-management commentary (Q2 and Q3 2025).